Concise Simulator Passage for Section II-A

(revision 2, audited against the code; supersedes the earlier sheet)

I. Methodology

A. Methane Leak Simulator

CH4-T places one source of unknown grid cell $c^* \in \{1, \dots, 64^2\}$ and rate q (10–500 $\mathrm{kg\,h^{-1}}$, log-uniform, spanning the 100 $\mathrm{kg\,h^{-1}}$ super-emitter threshold) on a 500×500 m site watched by $S \in \{4, \dots, 12\}$ masts at uniformly random positions. Each scenario comprises T=30 sequential snapshots: the wind evolves between snapshots, but each reading is an instantaneous steady-state plume evaluation, so no plume travel time or within-sample averaging is modelled and no result depends on the sampling cadence. Transport follows the standard Gaussian plume with published Pasquill–Gifford/Briggs dispersion, one stability class per scenario drawn uniformly from the six classes. In plume-frame coordinates (downwind distance x, crosswind offset y_{cw}, sensor height z_r, release height H), the concentration per unit rate is

$$g = \frac{1}{3600}\,\frac{1}{2\pi U \sigma_y(x)\sigma_z(x)}\, e^{-\frac{y_{\mathrm{cw}}^2}{2\sigma_y^2}} \left(e^{-\frac{(z_r-H)^2}{2\sigma_z^2}} + e^{-\frac{(z_r+H)^2}{2\sigma_z^2}} \right), \tag{1}$$

converted to ppm enhancement (1 ppm CH_4 = $6.55 \times 10^{-7}\,\mathrm{kg\,m^{-3}}$), with $g = 0$ upwind ($x \le 0$); U is the wind speed and σ_y, σ_z are the stability-dependent dispersion widths. The wind itself meanders: each scenario draws a mean speed (1–8 $\mathrm{m\,s^{-1}}$, log-uniform) and direction, modulated between snapshots by an autoregressive wobble of 20–40° in direction and 10–30% in speed, so the plume sweeps across the masts over the scenario. Stacking all sensors and snapshots, a leak at cell c produces the fingerprint $g_c(w) \in \mathbb{R}^{ST}$ under wind sequence w, and readings are

$$y \;=\; q\, g_{c^*}(w) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2 I), \tag{2}$$

with per-scenario noise scale $\sigma \in [0.2, 3.0]$ ppm (log-uniform, spanning low-cost to mid-grade sensors) and each reading independently missing with probability up to 0.2; all sums below run over the retained readings.

Wind uncertainty is built into the data. Readings are generated under the true wind w, but the dataset records only the measurement $\hat{w}$, corrupted independently at every step t:

$$\begin{aligned} \hat{\theta}_t &= (\theta_t + \delta_t) \bmod 360^\circ, & \delta_t &\sim \mathcal{N}(0, \sigma_\theta^2), \\ \hat{s}_t &= s_t\, e^{\eta_t}, & \eta_t &\sim \mathcal{N}(0, \sigma_s^2), \end{aligned} \tag{3}$$

with $\sigma_\theta = 10^\circ$ and $\sigma_s = 0.10$, the accuracy class of inexpensive monitoring anemometers. *Under the measured-wind condition used for all deployment results, no model or physics image at any stage observes the true wind*; the laboratory-limit audit of Section III-A deliberately supplies the exact wind as an upper bound on what is achievable.

What changed in revision 2, and why.

- **Time steps.** The earlier sheet said "T=30 one-minute steps." The code implements neither one-minute steps nor the "30 × 10-s averages" of the source docstring: each step is an independent instantaneous steady-plume evaluation with i.i.d. noise, and the stored `times` vector is dimensionless. Under a meandering wind a true one-minute average would *not* equal the instantaneous plume, so claiming a duration asserts physics the simulator does not implement. The wording above states what is actually modelled and keeps the cadence free.
- **The no-true-wind sentence is now scoped.** Stated absolutely, it contradicts Section III-A, whose laboratory limit is defined by exact wind and whose runs do feed the true wind to the model. Scoping it to the measured-wind condition removes the contradiction without weakening the claim that matters.
- **Retained from revision 1:** the ppm conversion (Eq. (1) was previously in $\mathrm{kg\,m^{-3}}$ while Eq. (2) is in ppm), the $g = 0$ upwind guard, the wind-meander specification, the stability draw, mast placement, the noise range, and the missing-data clause.
- **Verified, no change needed:** every quantitative range above was checked against both the code and the shipped data (rates 10.007–499.97 $\mathrm{kg\,h^{-1}}$; noise 0.200–3.000 ppm; realized dropout 0.099–0.102; stability classes 0.154–0.181 each; anemometer error 9.984° and 0.1000 in log-speed, independent across steps).
- **Consider dropping** the parenthetical "normalization checked to $\sim 10^{-12}$": the posterior is normalized by `logsumexp` before that residual is measured, so it reports float round-off, not quadrature accuracy. An independent 2×10^6-node trapezoid agrees with the shipped quadrature to a total variation of 6×10^{-14} — cite that instead.